LEAN KURT ESCANO

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL
Backend: Django, Django REST Framework, FastApi, PostgreSQL, MySQL, Spring Boot
Frontend & Mobile: React.js, React Native (Expo), TypeScript, NativeWind, Tailwind CSS, Zustand
Tools & DevOps/Cloud & Services: Linux/Unix Shell (Bash), Docker, Git , GitHub, CI/CD (GitHub Actions), OpenAI ,Vscode, Firebase, Postman, Render, Netlify, Pinecone, Hugging Face
Certifications: Meta Frontend Developer, Meta Backend Developer, Meta Database Engineer, Meta Full Stack Software Engineer - Professional Certificates(Coursera)

RELEVANT EXPERIENCE

Cloud Infrastructure & Disaster Recovery (Freelance)

Junior DevOps Engineer | September 2026 – Present

- Deployed a **backup website** connected to a separate **Supabase** project to provide an alternative environment for service recovery.
- Migrated financial application data and replicated the **PostgreSQL database schema** to a separate Supabase instance, preserving table structures, relationships, and constraints.
- Monitored cloud infrastructure and application health to identify operational issues and support system availability.

DTI LAGUNA ETAAS – Digital Trade & Services Platform (Laguna University × DTI Laguna Partnership)

Full Stack Developer | October 2025 – December 2025

- Engineered a full-stack marketplace connecting **30+ Laguna-based MSMEs** with customers, featuring product discovery, cart/checkout, order tracking, and multi-vendor dashboards.
- Developed a **multi-tenant Agentic RAG backend** using **FastAPI**, **Pinecone**, and **Gemini embeddings**, enabling each store owner to have a dedicated AI chatbot that retrieves answers exclusively from their store's product and business data, achieving **92% query-routing accuracy.**
- Integrated **Hugging Face Cross-Encoder reranking**, improving retrieval relevance by **35%** and reducing irrelevant chatbot responses from **18%** to **6%.**
- Built automated product-ingestion pipelines for embedding generation and vendor-isolated vector indexing, with asynchronous backend components designed to support **100+ concurrent users.**
- Integrated Firebase Authentication and Cloudinary for user authentication and media delivery, reducing mobile image load times by **60%.**

Commission Accounting System (Freelance)

Full-stack web application designed in collaboration with a real estate company to streamline commission
Full Stack Developer | January 2024 - March 2024

- *Built a scalable commission computation system used by **30+ agents** for real-time commission tracking.*
- *Implemented hierarchical calculations with custom rules, reducing manual computation errors by **95%**.*
- *Optimized **PostgreSQL** indexing, reducing hierarchical commission queries from **5.1s** to **0.9s** for real-time financial reporting*
- *Built **Django** RESTful APIs serving agent dashboards with performance analytics, decreasing monthly report generation from 3 hours to 15 minutes*

PROJECTS

COMPLAINTS AND FEEDBACK MANAGEMENT SYSTEM WITH AN INTEGRATED CHATBOT(FAQs)

Capstone Project | Laguna University
Mobile App: **https://play.google.com/store/apps/details?id=com.maryapp.user**
Web Platform: **https://www.cfms-stamaria.com/officials-login**

- Developed **Mary App**, a mobile complaint-reporting platform serving **25 barangays** in **Santa Maria, Laguna**, using **React Native (Expo)**, **NativeWind** and **TanStack Query**, alongside a **React** and **FastAPI** web dashboard for barangay personnel to manage and resolve complaints.
- Built an **OpenAI-powered emergency classification** and **routing system** to identify urgent complaints, direct users to the emergency
- interface, and classify incidents for appropriate response units, including **PNP**, **MDRRMO**, and **BFP**.
- Engineered a **hybrid AI-powered complaint deduplication pipeline** combining **Pinecone vector embeddings**, **semantic similarity**, **category-specific geographic radius filtering** using the **Haversine formula**, and **LLM verification** to identify duplicate reports of existing incidents.
- Implemented **Celery** background workers with **Redis** to process complaint-matching and AI verification tasks asynchronously, keeping API requests responsive.
- Developed a **Retrieval-Augmented Generation (RAG) chatbot** using **OpenAI models** and **Pinecone**, enabling administrators to upload **PDF documents** as a knowledge source for AI-generated answers to community FAQs.

QualitySense

Full Stack Web Application for Data Quality Analysis
*Deployed Project Link(Demo) : **https://qualitysense.netlify.app/***

- *Built a data quality engine capable of processing datasets up to **50K+ rows**, detecting nulls, outliers, and categorical anomalies using **pandas**.*
- *Engineered a **Django** backend that improved API response times by **30%** through optimized queries and caching strategies.*
- *Developed an AI-powered Data Quality Summary using **Gemini**, increasing user clarity on dataset issues.*
- *Implemented secure authentication using **JWT** + **Google OAuth 2.0**, eliminating unauthorized access events.*
- *Deployed responsive front-end UI with efficient state management for real-time dataset insights.*

EDUCATION

Bachelor of Science in Information Techology
Laguna University • 2023 – Present

AWARDS

Proficiency in Web Development – June 2025, Laguna University
Proficiency in Python Programming – June 2025, Laguna University
Proficiency in Mobile Application Development – June 2025, Laguna University